

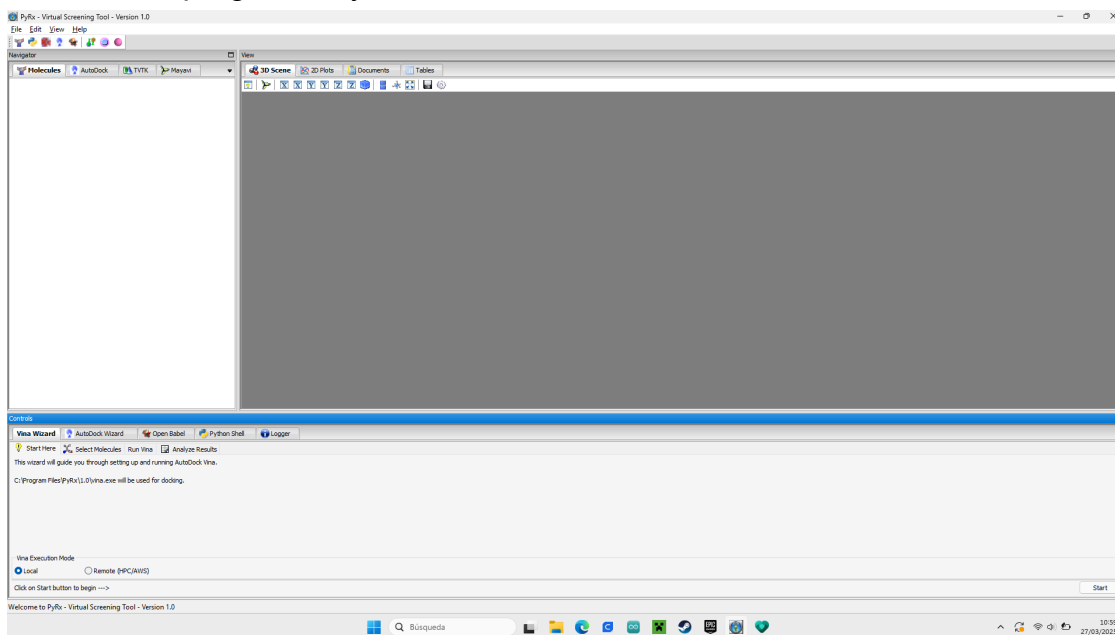
Guia PyRx

1. Descargar el programa Pyrx.
2. Acceder a bases de datos de proteínas y enfermedades relevantes para nuestro estudio:
 - PubChem: Buscar proteínas candidatas.
 - RCSB PDB: Consultar las enfermedades investigadas.
3. Descargar proteínas candidatas a antivirales, tanto en formato SDF como en 3D.

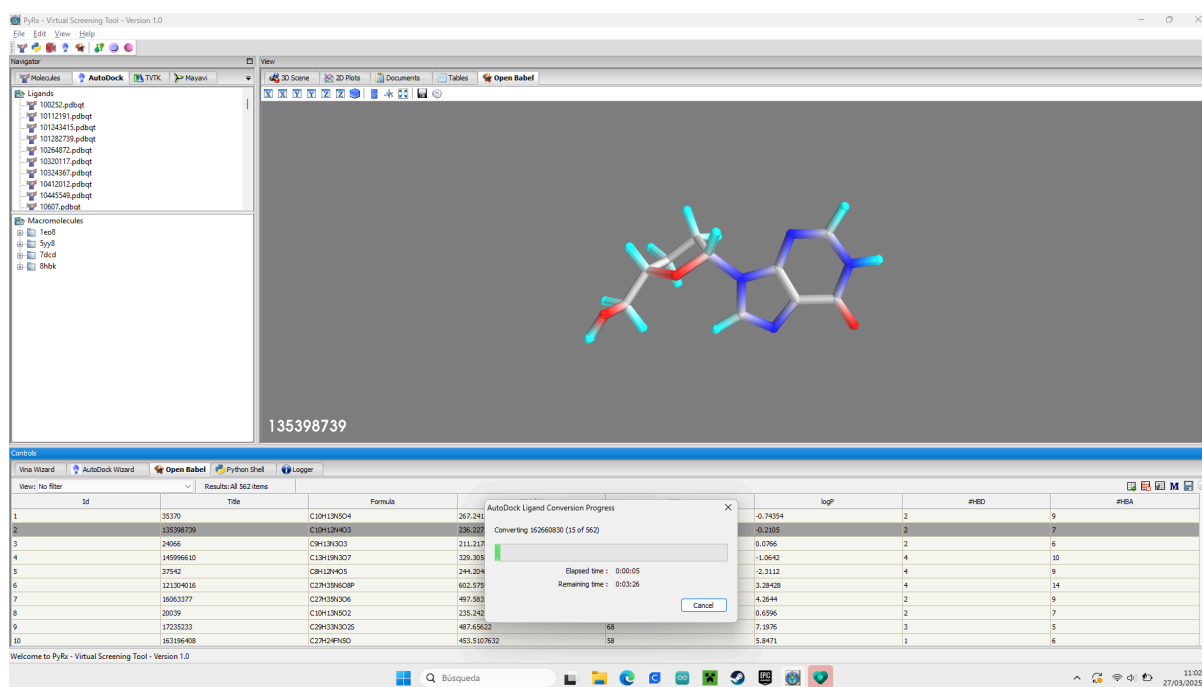
The screenshot shows the PubChem website interface. The search bar at the top contains the word 'antiviral'. Below the search bar, there are tabs for 'Compounds (952)', 'Substances (212)', 'Genes (114)', 'Proteins (50)', 'Pathways (40)', 'Cell Lines (2)', 'BioAssays (65,500)', 'Literature (191,715)', and 'Patents (34,637)'. The 'Compounds' tab is selected, showing 952 results. The first result is 'Oxymatrine; 16837-52-8; Ammothamine; Matrine N-oxide; Matrine oxide; ...'. The chemical structure is shown, along with its molecular formula $C_{15}H_{24}N_2O_2$ and molecular weight 264.36g/mol. The IUPAC Name is (1R,2R,9S,17S)-13-oxido-7-aza-13-azoniatetracyclo[7.7.1.0,2,7.0,13,17]heptadecan-6-one. The SMILES string is C1C[C@H]2[C@H]3CCCCN+4[C@H]3[C@H]2[C@H]1CCCC4CNC2=O[C@]3(D). The InChIKey is XVPBNOPNFXD-LHDFHYSN-N. The InChI string is InChI=1S/C15H24N2O2/c18-14-7-1-6-13-12-5-3-9-17(19)8-2-4-11(15)(12)17(10-16)(13)14/11-13,19(1)-10H(2)11-,12-,13-,15-,17(m)/s1. The Create Date is 2005-06-24. The compound is tagged by PubChem as D000890 - Anti-Infective Agents > D000998 - Antiviral Agents; D002317 - Cardiovascular Agents > D000889 - Anti-Arhythmia Agents.

This screenshot shows the same PubChem search results for 'antiviral'. A 'Descargas' (Downloads) dialog box is open in the top right corner, showing a list of files for download: 'PubChem_compound_text_antiviral_records (2).sdf' (0 kb - 2 types), 'PyRx-1.0-Setup.exe' (no se descarga habitualmente. Asegúrese de que confía en PyRx-1.0-Setup.exe antes de abrirlo.), and 'PubChem_compound_text_antiviral_records (2).sdf' (0 kb - 2 types). Below the dialog box, the 'DOWNLOAD' panel is visible, showing options for downloading the search results in various formats: CSV, JSON, XML, and ASNT. The 'Summary (Search Results)' section shows the first result, 'Oxymatrine; 16837-52-8; Ammothamine; Matrine N-oxide; Matrine oxide; ...', with its chemical structure and details. The second result is 'cytarabine; 147-94-4; Ara-C; Cytosine arabinoside; Aracytidine; ...'.

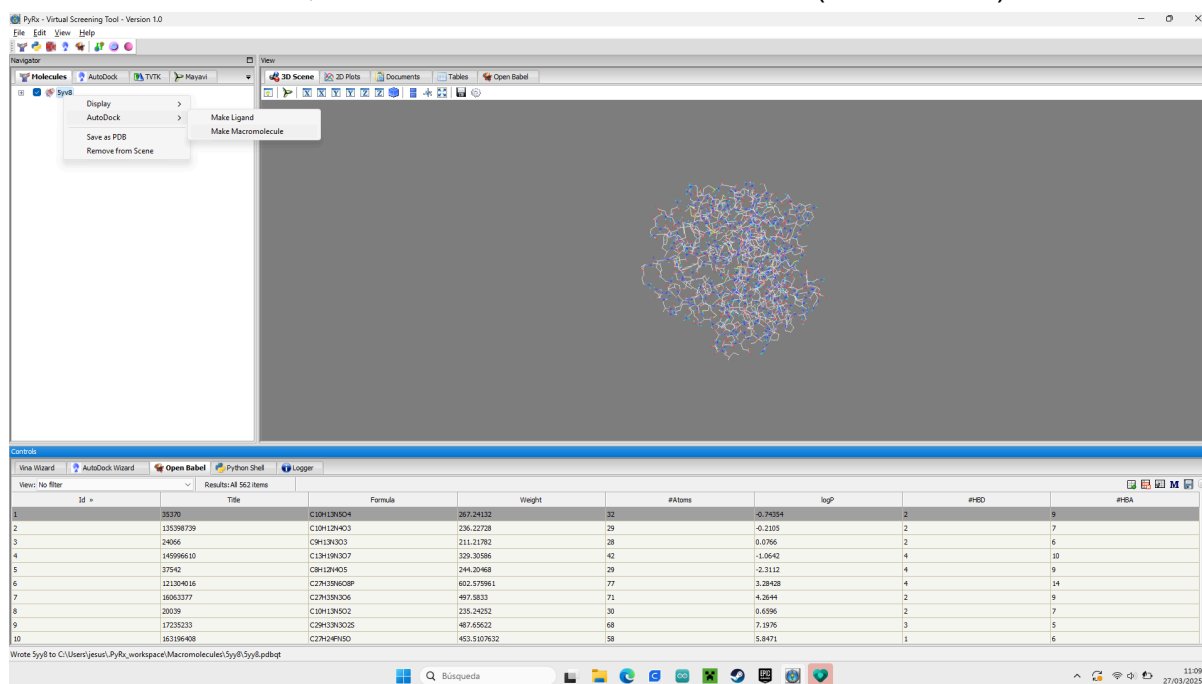
4. Abrir el programa PyRx.



5. Introducir todos los antivirales en Open Babel y convertirlos a formato "Autodock ligands".

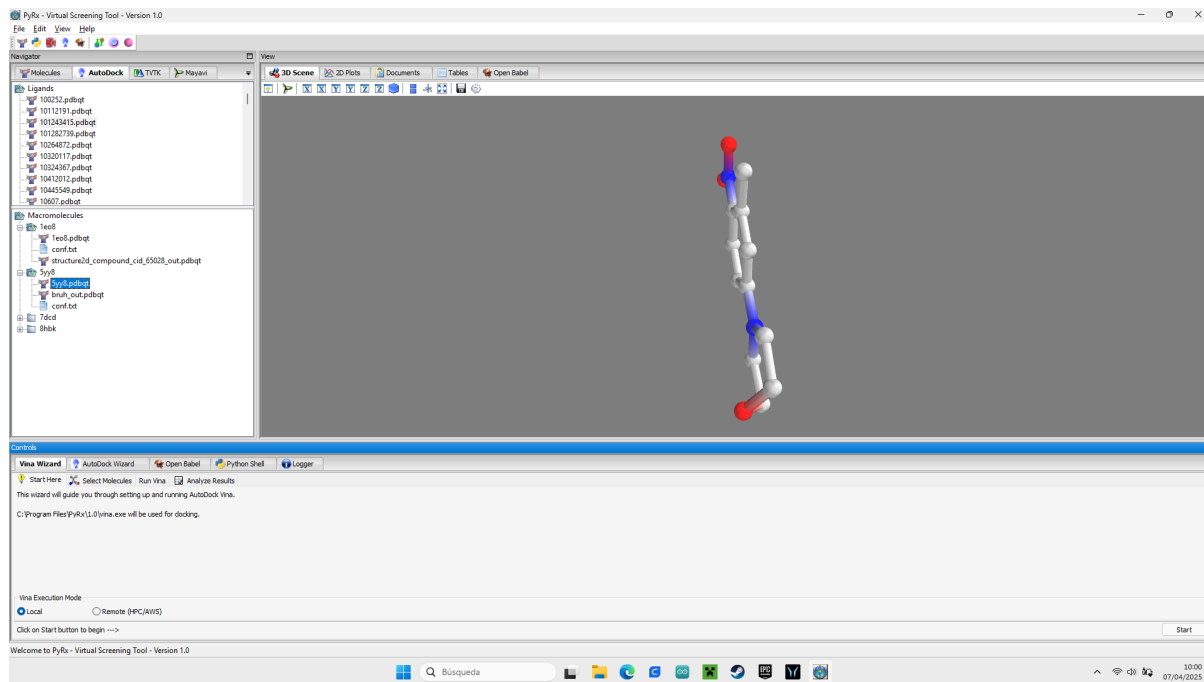
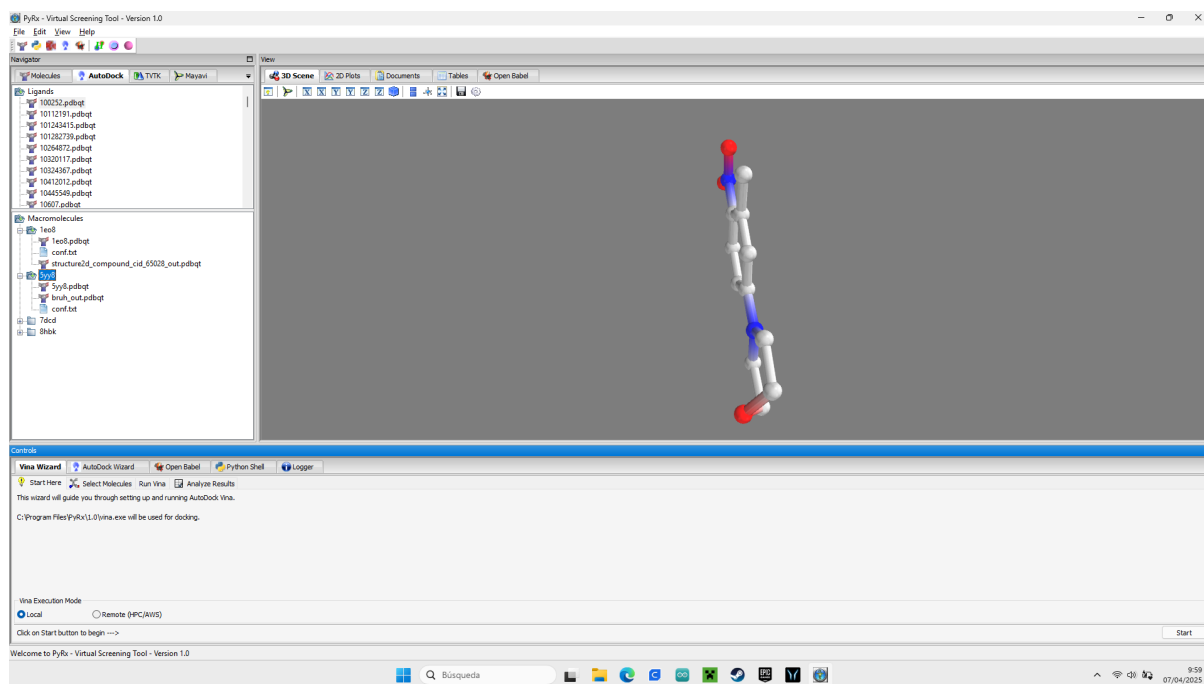


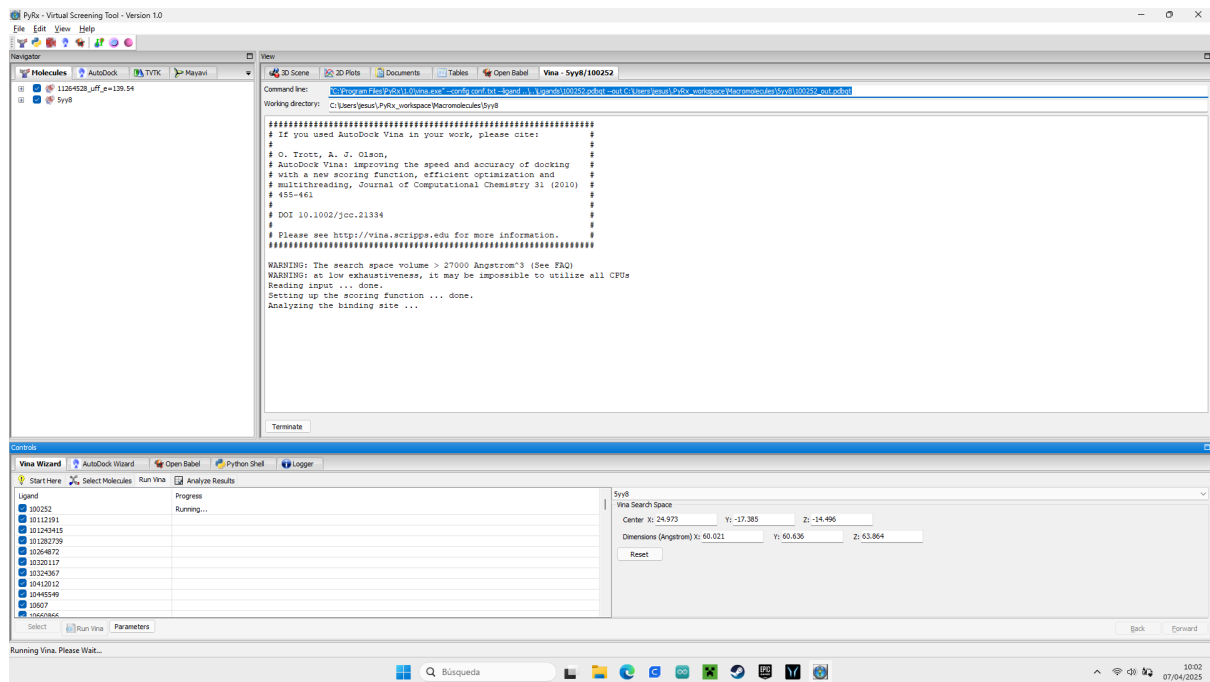
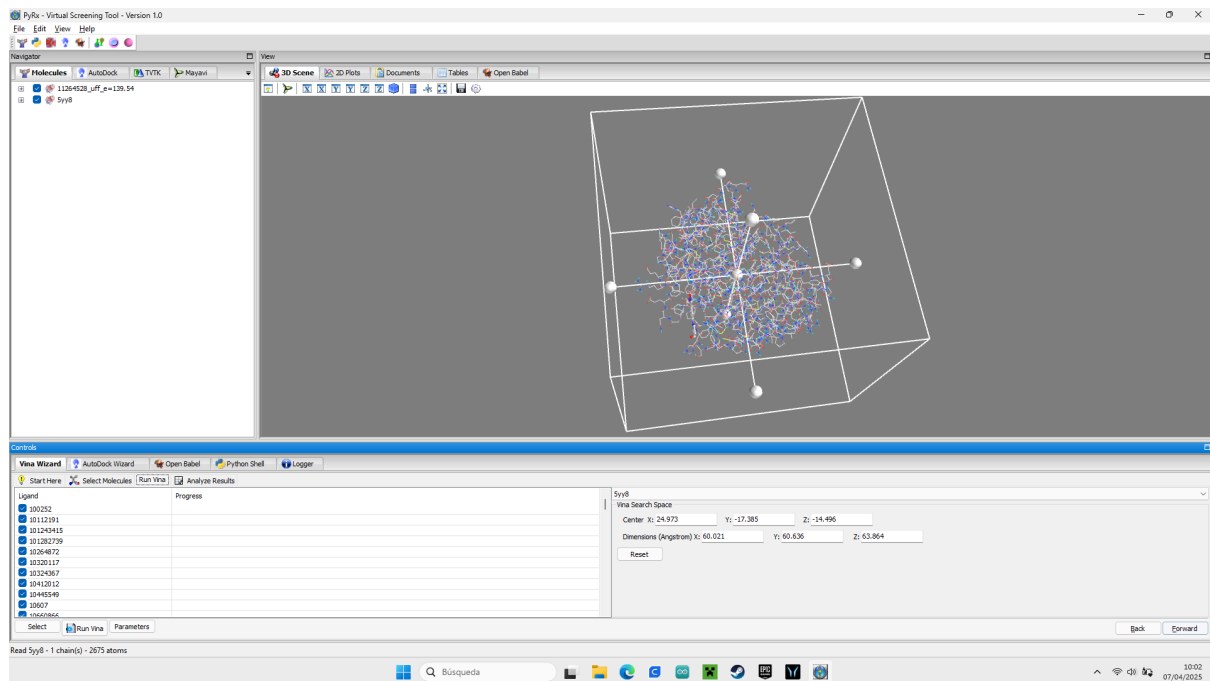
6. Descargar una proteína relacionada con la enfermedad desde el RCSB PDB en formato PDB, convertirla en una macromolécula (clic derecho).

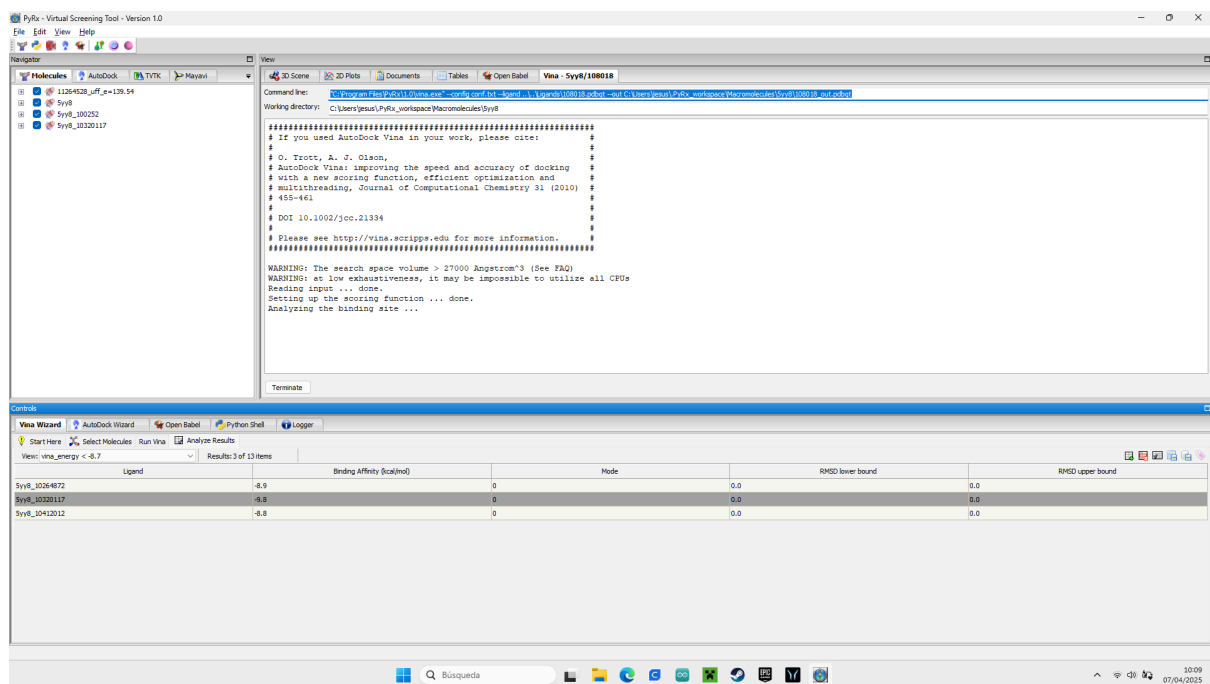


7. Abrir el Vina Wizard y realizar el docking con Autodock:

- Seleccionar la proteína convertida en macromolécula.
- En Vina, hacer clic en "Forward".
- Seleccionar todos los ligandos y volver a hacer clic en "Forward".
- Repetir el proceso y hacer clic en "Forward" nuevamente.







8. Analizar los resultados y enfocarse en obtener valores cercanos a -12 para optimizar el resultado.